MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLES, 90 THROUGH 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification sheet is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.

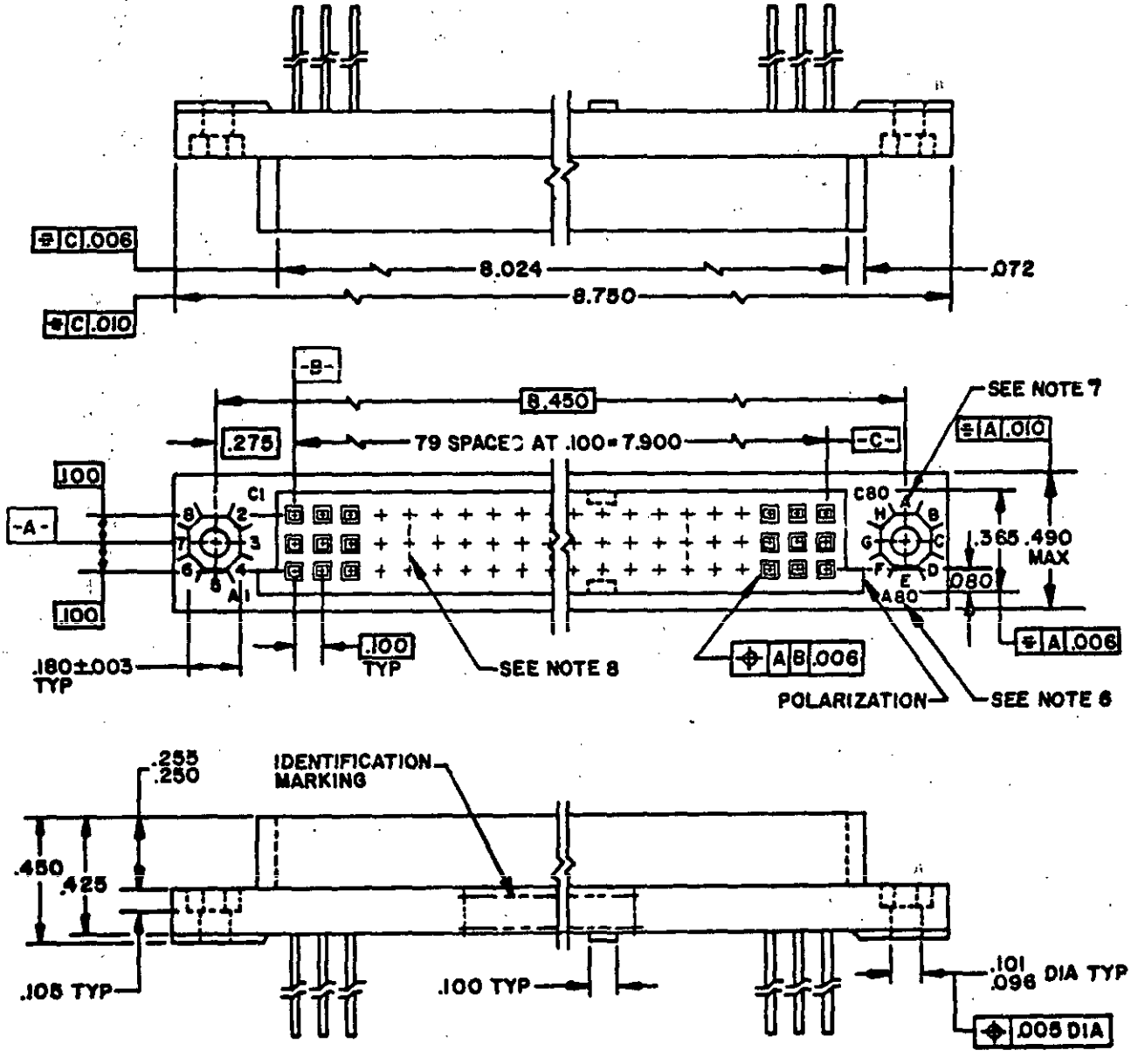
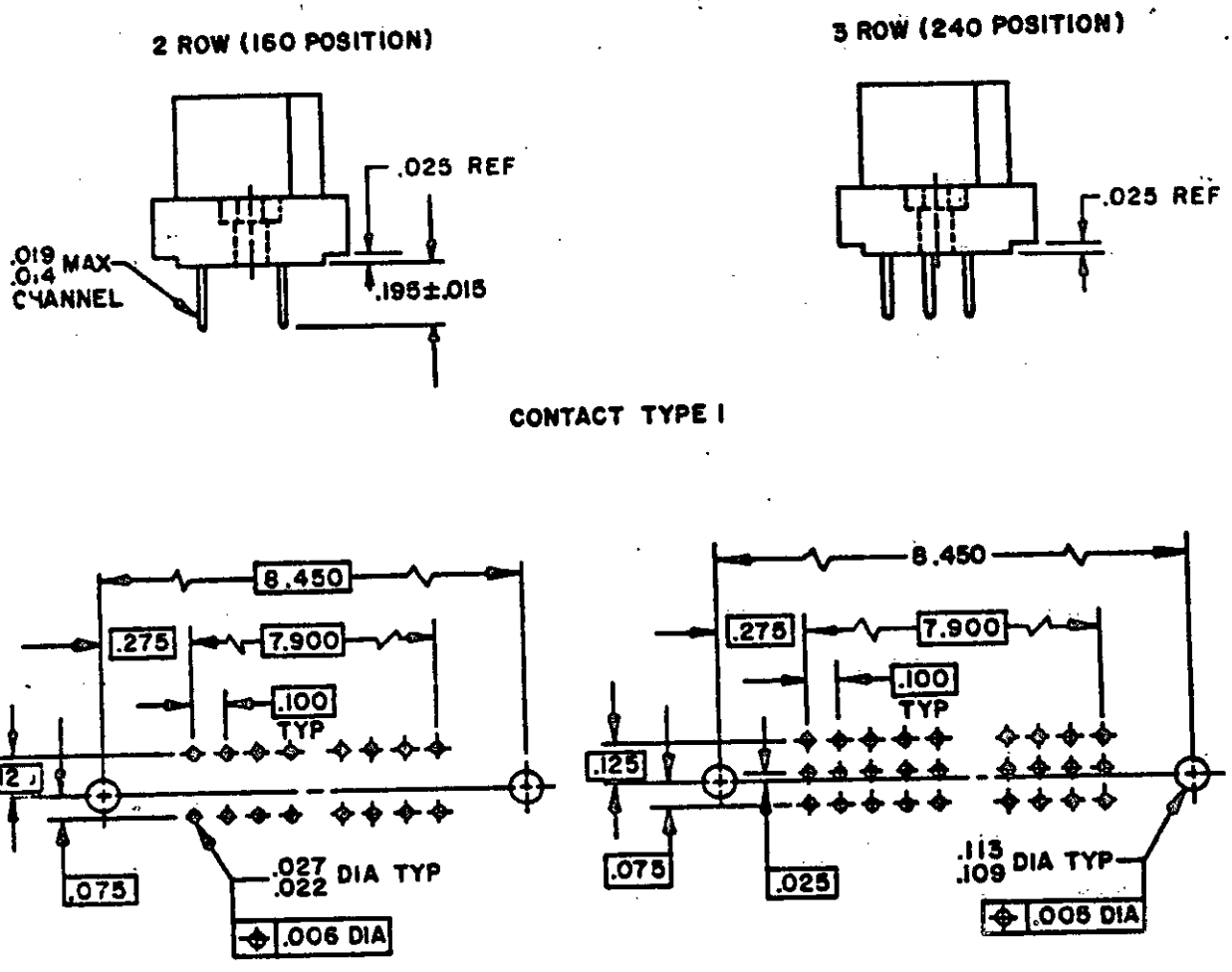


FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts.

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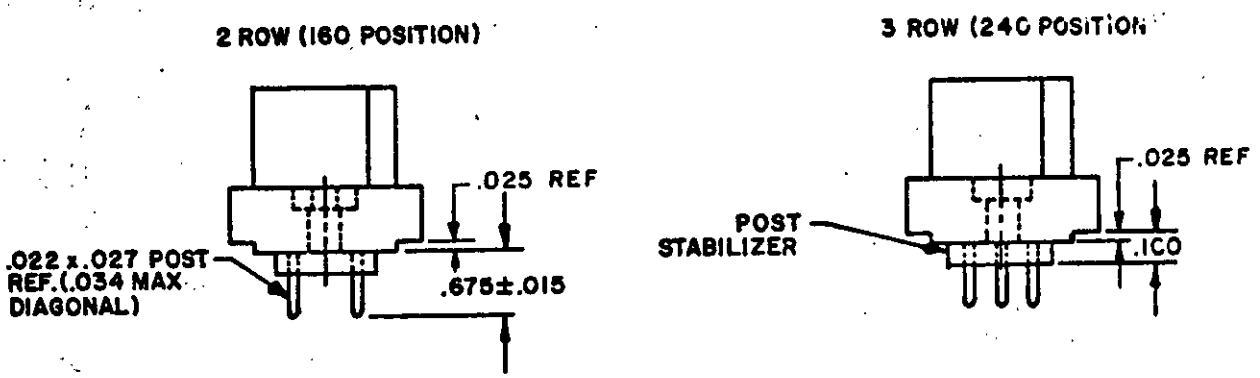
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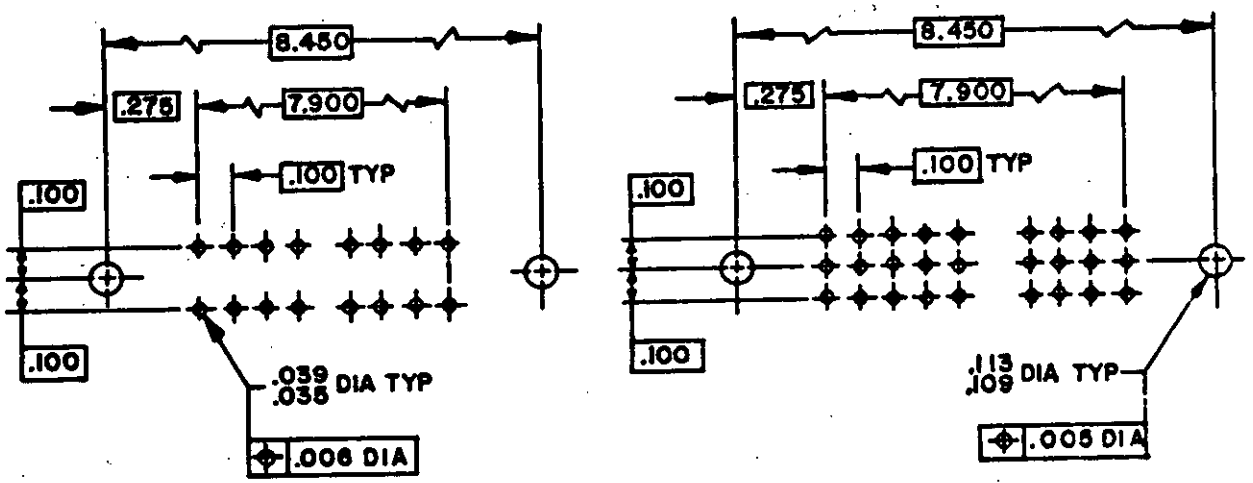
CONTACT TYPES AND RECOMMENDED CIRCUIT LAYOUTS

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

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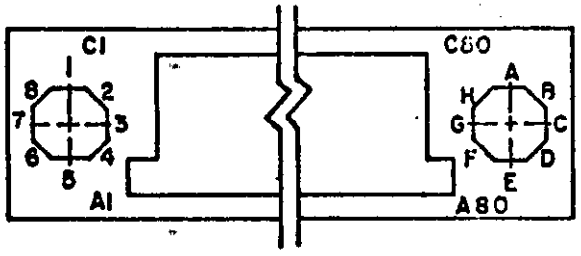
CONTACT TYPE 2



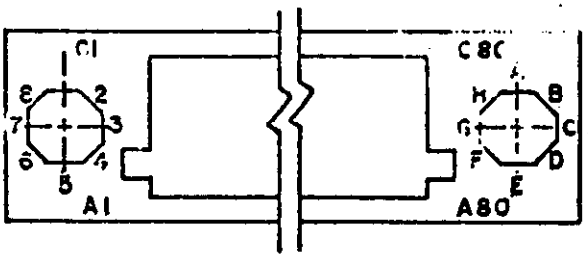
CONTACT TYPES AND RECOMMENDED CIRCUIT LAYOUTS

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

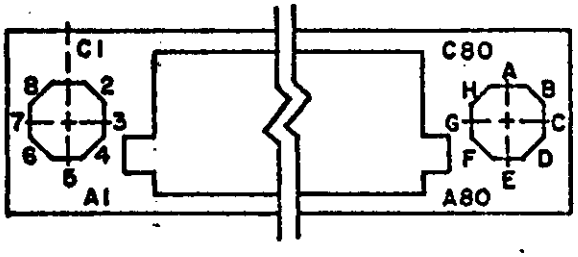
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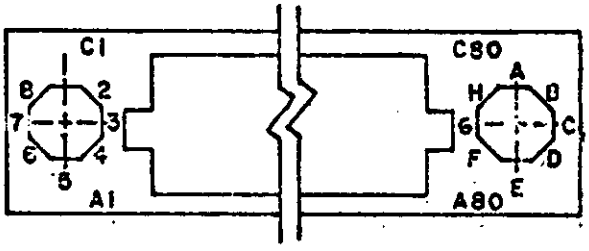
TYPE I



TYPE II



TYPE III



TYPE IV

POLARIZATION

Inch	MM	Inch	MM	Inch	MM
.003	.08	.039	.99	.255	6.48
.005	.13	.072	1.83	.275	6.99
.006	.15	.073	1.85	.365	9.27
.010	.25	.080	2.03	.425	10.80
.014	.36	.096	2.44	.450	11.43
.015	.38	.100	2.54	.490	12.45
.019	.48	.109	2.77	.675	17.15
.022	.56	.113	2.87	7.700	200.66
.027	.69	.180	4.57	8.024	203.81
.034	.86	.195	4.95	8.450	214.63
.035	.89	.250	6.35	8.750	222.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/129.
5. Use printed circuit boards with both the type 1 and type 2 receptacle.
6. Numbers indicating end cavities and letters indicating rows stamped this surface.
7. Location indicators embossed on surface.
8. Lines indicate every five cavities stamped on mating face except end cavities.

FIGURE 1. Connectors, kit receptacles (.100 spacing) - 160 and 240 contacts - Continued.

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TABLE I. Dash number; Polarization type, Contact type for 160 and 240 contact position.

Dash number	Number contacts	Type polarization	Type contact
01	160	I	1
02	160	I	2
03	160	II	1
04	160	II	2
05	160	III	1
06	160	III	2
07	160	IV	1
08	160	IV	2
09	240	I	1
10	240	I	2
11	240	II	1
12	240	II	2
13	240	III	1
14	240	III	2
15	240	IV	1
16	240	IV	2

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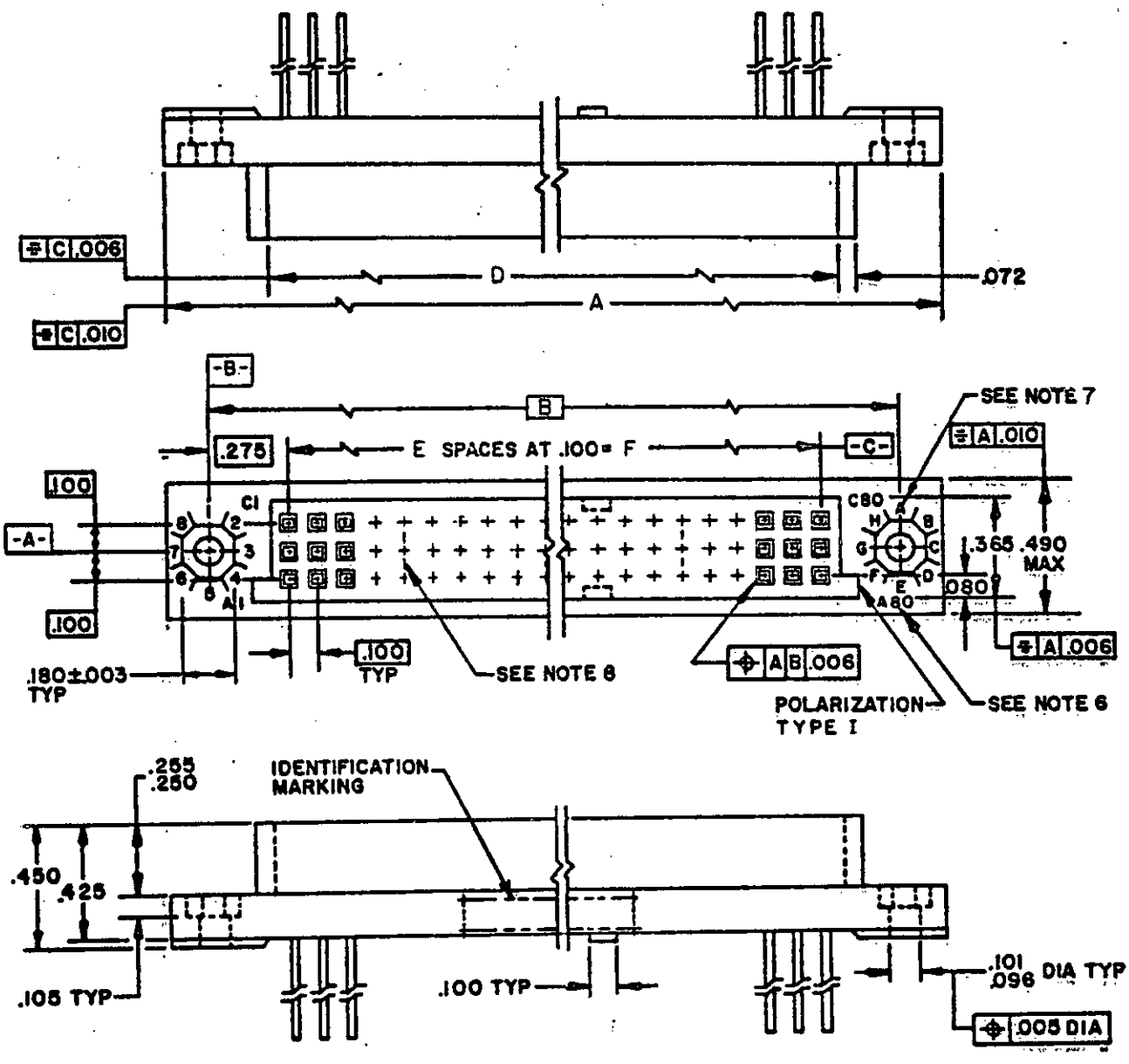
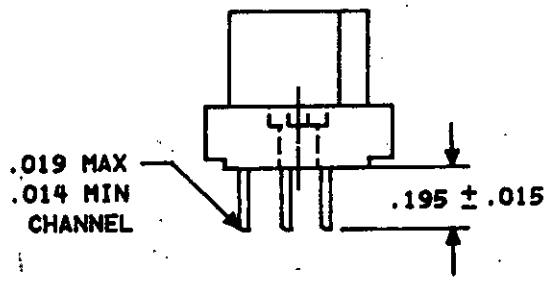
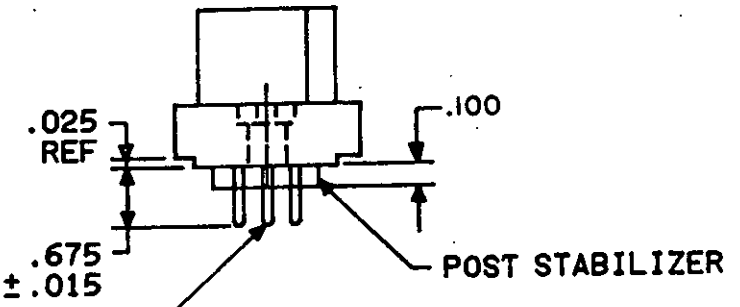


FIGURE 2. Connectors, receptacles (.100 spacing) 90 through 240 contact positions.

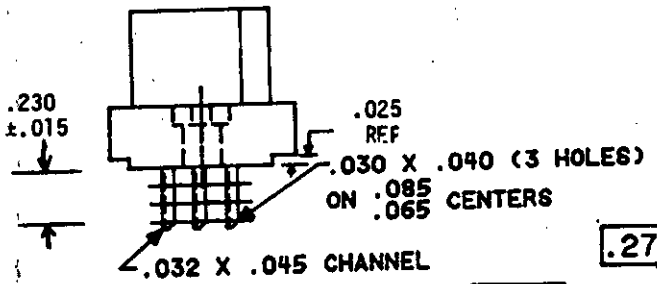
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FLOW SOLDER TYPE
CONTACT TYPE 1

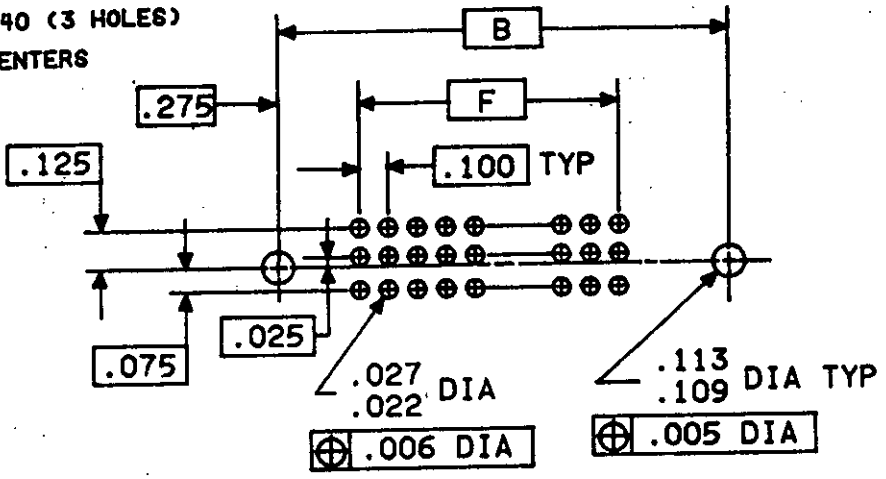


RACK MOUNT
WIRE WRAPPED TYPE
CONTACT TYPE 2



EYELET CONTACT
(SOLDER)
NO.26 AWG MAX

CONTACT TYPE 3



RECOMMENDED PRINTED CIRCUIT BOARD LAYOUT
CONNECTOR SIDE
FLOW SOLDER TYPE
CONTACT TYPE 1

INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.027	.69	.085	2.16
.006	.15	.030	.76	.100	2.54
.014	.36	.032	.81	.125	3.18
.015	.38	.034	.86	.195	4.95
.019	.48	.040	1.02	.675	17.14
.022	.56	.045	1.14		
.025	.64	.075	1.90		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Connectors, receptacles (.100 spacing) 90 through 240 contact positions - Continued.

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TABLE II. Dash numbers and dimensions for 90 through 240 contact positions.

Dash no.	No. of contacts	Type	A	B	D	E	F	Contact ident. no.
17 18 19	90	1 2 3	3.750 (95.25)	3.450 (87.63)	3.024 (76.81)	29	2.900 (73.66)	1-30
20 21 22	105	1 2 3	4.250 (107.95)	3.950 (100.33)	3.524 (89.51)	34	3.400 (86.36)	1-35
23 24 25	120	1 2 3	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)	1-40
26 27 28	135	1 2 3	5.250 (133.35)	4.950 (125.73)	4.524 (115.57)	44	4.400 (111.76)	1-45
29 30 31	150	1 2 3	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)	1-50
32 33 34	165	1 2 3	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)	1-55
35 36 37	180	1 2 3	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)	1-60
38 39 40	195	1 2 3	7.250 (184.15)	6.950 (176.53)	6.524 (166.37)	64	6.400 (162.56)	1-65
41 42 43	210	1 2 3	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)	1-70
44 45 46	240	1 2 3	8.750 (222.25)	8.450 (214.63)	8.024 (203.81)	79	7.900 (200.66)	1-80

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See Figures 1 and 2, and TABLE I and II.

Material: In accordance with MIL-C-55302, except body and stabilizer material type GPT-30F per MIL-M-24519.

Plating: Type 1, 2, and 3 shall be nickel plated per QQ-N-290, Class 2, 30 to 150 microinches thick. Over the nickel plate, the mating engagement area of Type 1 and 3 for a minimum length of .200 inch including the retention tabs and the entire contact Type 2 shall be gold palted per MIL-G-45204, Type II, grade C, class 1. Over the nickel plate, the solder tail area for Type 1 and 3 shall be tin-lead plated for a minimum length of .215 inch and .270 inch respectively per MIL-T-10727, Type I, 100 microinches minumum.

Contact identification: See figure 1 and 2, and tables I and II

Current rating: 3.0 amperes maximum per contact; 2.25 amperes continuous per connector at room ambient.

Keying (see MIL-C-55302/31):

Dash numbers -01 through -16: Two keys, part number M55302/31-08, and two key mounting hardware, part number M55302/31-09, packaged with connectors.
Dash numbers -17 through -46: Key, part number M55302/31-04, with .086-56 UNC-2A mounting screw is recommended.

Replacement contacts: See MIL-C-55302/32.

Contact removal tool: See MIL-I-81969/9.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 0.25; the withdrawal force in pounds shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohm.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude: 200 volts rms.

Resistance to test probe damage: The connector shall meet the requirements of MIL-C-55302/30.

Contact separation force: The connector shall meet the requirements of MIL-C-55302/29.

Oversized pin exclusion: .0465 diameter pin.

Part number: M55302/130 (and dash number from table I for 160 and 240 contact positions).
M55302/130 (and dash number from table II for 90 through 240 contact positions).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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